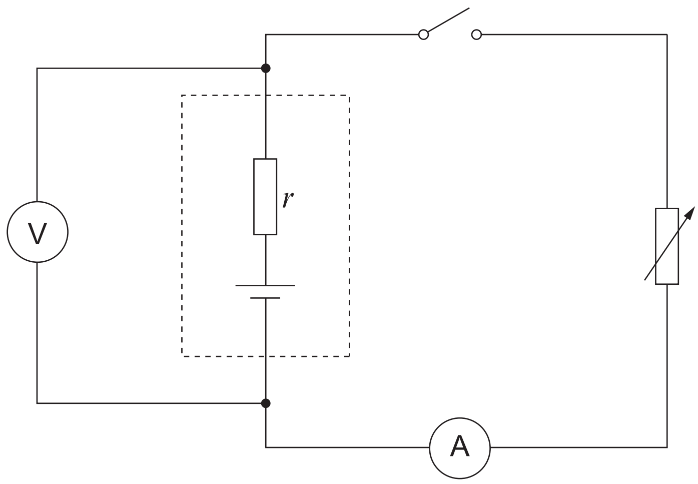
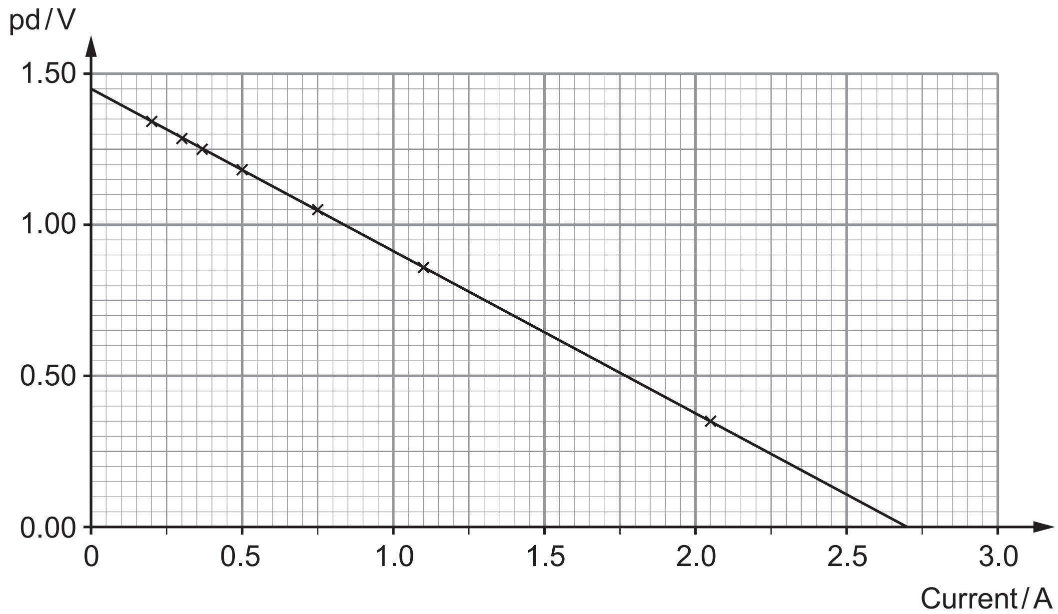
Examiner only

Answer **all** questions.

1. Anwen sets up the circuit shown in order to investigate a cell. The variable resistor is adjusted and readings are obtained from the voltmeter and ammeter.



The readings are plotted as shown.



(a) The manufacturer claims that the emf of the cell is 1.50 V. Explain, in terms of energy, what this statement means. [2]

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Examiner only

(b) An equation applying to a cell of emf, E, and internal resistance, r, is:

$$V = E - Ir$$

(i) Explain how Anwen's graph is in good agreement with this equation. [2]

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(ii) Calculate the gradient of the line and hence determine a value for the cell's internal resistance. [2]

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(iii) Evaluate the manufacturer's claim that the emf is 1.50 V. [1]

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(c) (i) Use the graph to determine the greatest current the cell can supply. [1]

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(ii) State the resistance of the variable resistor when the current is at its maximum. [1]

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